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Formation mechanisms of welding defects induced by wire offset in laser welding with filler wire: Effect of energy density distribution and molten pool dynamics



Zihan Li ^a, Fang Li ^{a b c} , Zhengxian Li ^d, Yilin Chen ^d, Gang Ruan ^a, Jiacheng Huang ^a, Lin Wang ^a, Chen Shen ^{a b c}, Xueming Hua ^{a b c}

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Highlights

- Positive longitudinal offset induces initial underfill, while negative offset triggers droplet transfer and humping.

Loading [MathJax]/jax/output/SVG/jax.js utilizes the welding process by absorbing 7.11 % of laser energy and contracting keyhole by 20 %.

- Transverse wire offset causes asymmetric energy density and fluid flow, leading to undercut defects on wire offset side.
- Transverse wire offset ($|\Delta y| > 0.5 \text{ mm}$) is more detrimental to weld morphology than longitudinal offset ($\Delta x < -1 \text{ mm} / \Delta x > 2 \text{ mm}$).

Abstract

Filler wire feeding is an effective strategy for optimizing weld morphology in laser welding. However, this approach significantly increases process complexity, where even a minor offset of the wire tip may induce welding defects. Using an integrated experimental and numerical approach, this study reveals the formation mechanisms of defects induced by wire offset in the longitudinal (Δx) and transverse (Δy) directions. The results indicate that the transverse wire offset has a more severe effect on weld morphology than the longitudinal wire offset. A transverse wire offset ($\Delta y = 0.5 \text{ mm}$) leads to asymmetric energy density and fluid flow, causing insufficient refilling of the keyhole and the formation of undercut defects on the wire offset side. When a longitudinal wire offset is introduced, two typical mass transfer modes are identified: the stable liquid bridge transfer mode and the unstable droplet transfer mode. With a positive offset ($\Delta x = 3 \text{ mm}$), underfill occurs at the initial stage due to the delay of wire feeding. Conversely, a negative longitudinal wire offset ($\Delta x = -2 \text{ mm}$) promotes a droplet transfer mode, where the periodic droplets disrupt the stability of the molten pool and result in humping defects. This study establishes a stable process window for wire offsets and elucidates that welding defects are governed by the coupling of asymmetric energy density distribution and molten pool dynamics, providing a scientific basis for optimizing the wire feeding process in laser welding.



Keywords

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Laser welding with filler wire; Wire offset; Welding defect; Energy density distribution; Molten pool dynamics

1. Introduction

Owing to the advantages of high energy density [1], high welding efficiency [2], and narrow heat-affected zone [3], laser welding has become one of the most widely utilized joining methods in many industries, such as aerospace [4] and automotive manufacturing [5]. However, the laser welding process is highly sensitive to assembly conditions, in which even a minor misalignment can lead to imperfect weld morphology or welding defects [6,7]. Therefore, the inherent assembly misalignment in the manufacturing process due to thermal distortion and tolerances gives rise to a significant challenge [8], which severely limits the reliability of the laser welding process, especially under precise welding conditions [9]. To mitigate these stringent assembly requirements, filler wire is widely introduced into the laser welding process [10]. This process not only compensates for assembly misalignments but also enables metallurgical modification and increases weld reinforcement, thereby improving the mechanical performance and ensuring the structural integrity of weld joints [11,12].

While the filler wire feeding provides several advantages, it also introduces significant process complexity [13]. During the laser welding process with filler wire, dynamic disturbances such as friction and robotic vibrations in the wire feeding conduit cause the filler wire to deviate, thereby resulting in an offset between the filler wire and the laser beam [14]. Such an offset disrupts the coupling between the laser beam, filler wire, and molten pool, thereby leading to welding defects [15]. Huang et al. [16] elucidated the effect of the longitudinal wire offset on the weld morphology. The wire offset determined the energy propagation and force analysis in the molten pool, thereby governing the mass transfer mode. When the droplet transfer mode dominated, the laser welding process was disrupted, resulting in undesirable weld morphology. Conversely, a liquid bridge transfer mode led to a more stable process and a well-formed weld seam. Zhang et al. [13] investigated the asymmetric fusion zone in laser welding with a deviated filler wire. As a transverse wire offset was introduced, a significantly asymmetric temperature distribution was observed in the molten pool. This asymmetry led to a "golf club-like" molten pool, where the weld width was narrower on the wire offset side. While these experimental studies [13,16] have correlated the wire offset with weld morphology, the underlying mechanism—particularly how this wire offset influences energy density distribution and molten pool dynamics—remains unclear. This knowledge gap primarily arises from the limitations of in-situ monitoring techniques in capturing these highly

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To overcome this experimental limitation, computational fluid dynamics (CFD) simulation has emerged as a powerful tool for revealing the underlying mechanisms governing the laser welding process. The high-fidelity CFD models developed by Meng et al. [19] and Duan et al. [20] have successfully simulated the laser welding process with filler wire, revealing that the molten filler wire established a liquid metal bridge to the molten pool. Furthermore, the filler wire feeding significantly modulated molten pool dynamics by introducing additional mass and momentum, thereby facilitating the fluid flow. Although the previous studies [19,20] have established a solid theoretical foundation for the numerical simulation of the laser welding process with filler wire, these works primarily focused on the advantages of the filler wire feeding under ideal conditions. However, under the non-ideal conditions in practical manufacturing, the process is susceptible to wire offsets, which destabilize the laser welding process and induce welding defects [13,16]. Therefore, it is necessary to elucidate a comprehensive causative pathway that links wire offset to non-uniform energy density distribution, which induces unbalanced molten pool dynamics and ultimately results in welding defects.

In this study, we employ an integrated approach of experimental analysis and numerical modeling to investigate the formation mechanism of defects induced by wire offset during the laser welding process with filler wire. The work elucidates the mechanistic role of a longitudinal wire offset in inducing the transition between the liquid bridge mode and droplet transfer mode, and its ultimate effect on process stability. Furthermore, we examine how a transverse wire offset disrupts the symmetry of the energy density distribution, and establish how this asymmetry governs the fluid flow field and the formation of undercut defects. These findings not only elucidate the underlying formation mechanism of defects induced by wire offsets but also provide a guidance for improving wire feeding stability and weld quality in laser welding.

2. Experimental procedures

2.1. Materials

Laser welding was conducted on 1.4 mm thick 22MnB5 sheets. A filler wire with a diameter of 1.0 mm was introduced in laser welding. The chemical compositions of the base metal and the filler wire are presented in Table 1.

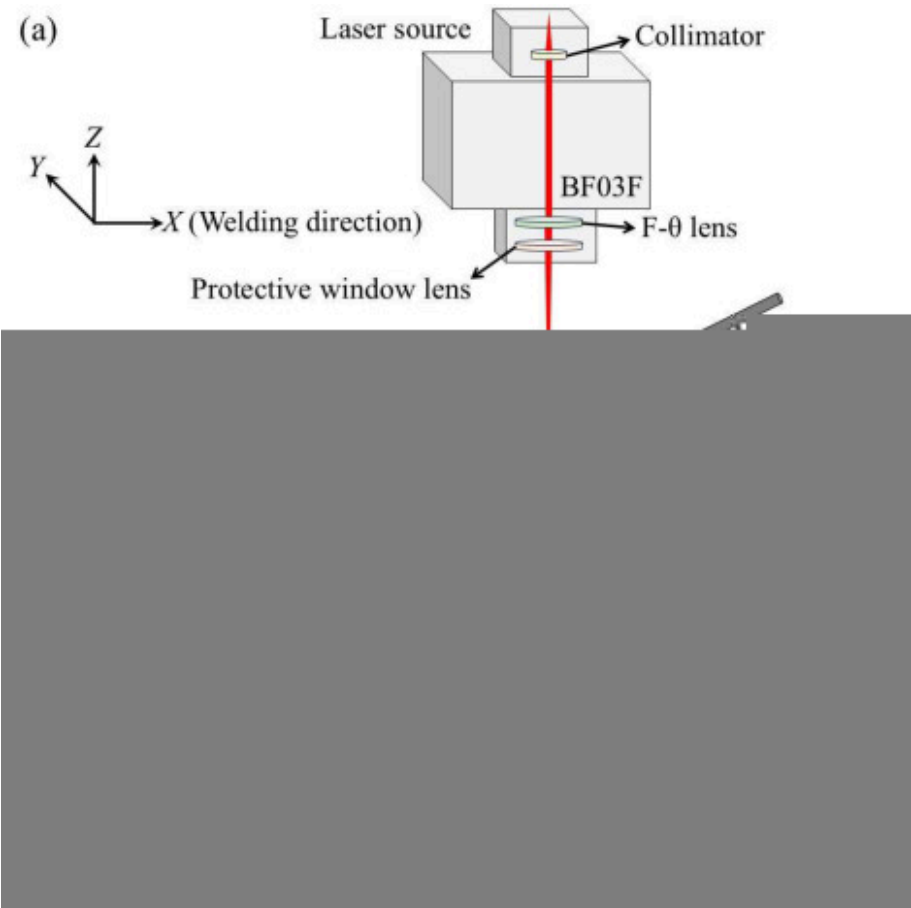
Table 1. Chemical compositions of the base metal and the filler wire (wt. %).

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	C	Si	Mn	B	Cr	Al	Ti	P	Fe
22MnB5	0.214	0.25	1.18	0.0026	0.23	0.029	0.03	0.011	Bal.
Filler wire	0.22	0.4	1.2	-	-	-	-	-	Bal.

2.2. Welding parameter

The experimental setup for the laser welding process is presented in Fig. 1, and Table 2 lists the welding parameters. During the laser welding process, the molten pool was monitored using a high-speed camera (Phantom VEO 710S) at 5000 frames per second, with illumination provided by a pulsed diode laser light source (CAVILUX® HF). After laser welding, a 3D camera (SICK Ruler XR 100) was used to measure the height of the weld seam.



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Fig. 1. Experimental setup. (a) Laser welding system. (b) 3D camera system. (c) High-speed camera system.

Table 2. Welding parameters.

Parameter	Value	Unit
Laser power	5000	W
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Parameter	Value	Unit
Defocus distance	0	mm
Welding speed	7	m/min
Wire feeding speed	1.5	m/min
Wire feeding angle	30	°
Wire feeding direction	Leading	-

In this study, the wire offsets were defined as the relative distance between the laser beam and the filler wire in the x and y directions, as shown in Fig. 1(a). The longitudinal wire offset (Δx) was set from -2 mm to 3 mm, while the transverse wire offset (Δy) was set as 0 mm, 0.25 mm, and 0.5 mm, as listed in Table 3. These wire offsets were determined based on the preliminary experiments. When the wire offset exceeded the designed range, the laser welding process failed to form a continuous weld seam. Additionally, to improve the experimental repeatability, three independent replicates were conducted for each wire offset in Table 3.

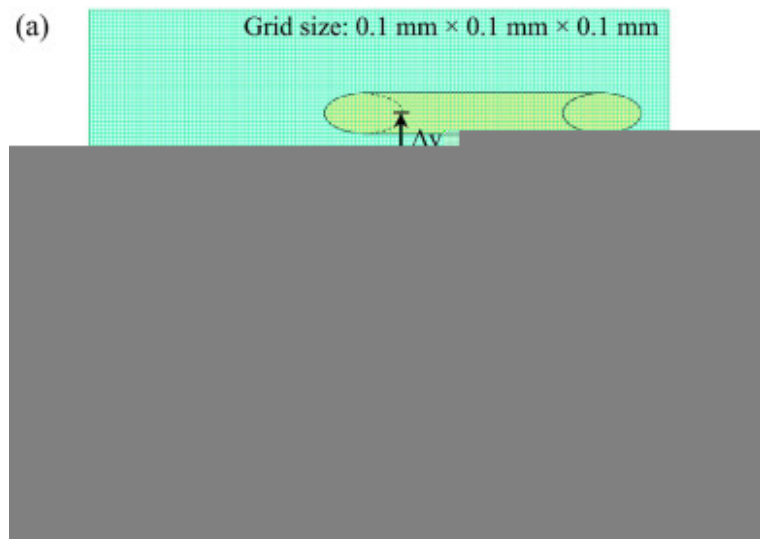
Table 3. Wire offsets used in the experiment.

Serial number	Δx / mm	Δy / mm
1–6	-2, -1, 0, 1, 2, 3	0
7–12	-2, -1, 0, 1, 2, 3	0.25
13–18	-2, -1, 0, 1, 2, 3	0.5

3. Numerical model

Loading [MathJax]/jax/output/SVG/jax.js and heat source model

Fig. 2 illustrates the three-dimensional computation domain in this study. The computation domain was 20 mm in length (x), 8 mm in width (y), and 5.4 mm in height (z). The spatial discretization of this domain was achieved using hexahedral grids. To determine the grid size of the numerical model, a grid independence study was performed. Three models with different grid sizes (0.14 mm, 0.1 mm, and 0.07 mm) were developed when $\Delta x = 0$ mm and $\Delta y = 0$ mm. For the model with the coarsest grids (0.14 mm), the numerical simulation failed to predict full sheet penetration and exhibited significantly different molten pool dynamics. In contrast, the simulated results of the models with 0.1 mm and 0.07 mm grids were similar and closely matched the experimental results. Based on a balance between computational accuracy and efficiency, uniform hexahedral grids of 0.1 mm were adopted in our model.



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Fig. 2. Computation domain of the numerical model. (a) Top view. (b) Side view.

As shown in Fig. 2(b), the computational domain was divided in the z direction into three distinct regions: a 1.4 mm thick sheet and two void regions. In this numerical model, the laser beam was assumed to be a stationary heat source, and the base metal moved at the welding speed in the reverse direction. The filler wire was introduced at $(17.06 + \Delta x$ mm, $4 + \Delta y$ mm, 4.9 mm), with its feeding speed and direction consistent with the welding parameters listed in Table 2. Since the filler wire and the base

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metal are both low carbon-manganese steels, they were assigned identical thermodynamic properties in the model. The thermodynamic properties of the base metal have been presented in the previous work [21].

A ray-tracing algorithm was employed to simulate a stationary laser heat source. The laser beam was divided into 52×52 sub-rays at the focus plane, and the energy of each sub-ray was as follows [22]:

$$q(x, y) = \begin{cases} \frac{3P_{laser}}{\pi r_q^2} \cdot \exp\left(-\frac{3(x-x_0)^2 + 3(y-y_0)^2}{r_q^2}\right) \cdot \left(\frac{2r_q}{52}\right)^2, & r < r_q \\ 0, & r \geq r_q \end{cases} \quad (1)$$

where P_{laser} is the laser power, (x_0, y_0) denotes the position of the stationary laser beam ($x_0 = 12$ mm, $y_0 = 4$ mm), and r_q represents the effective radius of the Gauss heat source.

3.2. Governing equations

The simulation was conducted using the commercial software Flow-3D to reproduce the complex physical phenomena in laser welding, including heat transfer and fluid flow. The mass (Eq. (2), [23]), momentum (Eq. (3), [23]), energy (Eq. (4), [23]), species (Eq. (5), [24]) conservation equations, and volume of fluid (Eq. (6), [23]) equation are as follows:

$$\frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v) = M_w \quad (2)$$

$$\begin{cases} \frac{\partial(\rho v)}{\partial t} + \rho v \cdot \nabla v = -\nabla P + \mu \nabla^2 v - \mu K v + \rho g + F_w \\ K = K_0 \frac{(1 - F_L)^2}{F_L + \delta} \end{cases} \quad (3)$$

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho v h) = \nabla \cdot (k \nabla T) + E_w \quad (4)$$

$$\frac{\partial \rho \Phi}{\partial t} + \nabla \cdot (\rho v \Phi) = \Phi_w \quad (5)$$

$$\frac{\partial \rho F}{\partial t} + \nabla \cdot (\rho v F) = 0 \quad (6)$$

where ρ denotes the density of the molten metal, v is the velocity vector, t represents the time, and M_w means the mass source term, which is provided by the molten filler wire. P is the pressure, μ indicates the dynamic viscosity, K is the Carman-Kozeny

coefficient, g represents the gravitational acceleration, F_W is the momentum source term, K_0 means the mushy zone constant, F_L is the fluid fraction, and δ denotes a small positive constant. h is the enthalpy, k indicates the thermal conductivity, T means the temperature, and E_W represents the heat source term. ϕ is the weight percentage of the filler wire, ϕ_W indicates the species source term caused by the molten filler wire, and F is the volume fraction.

3.3. Driving force and boundary conditions

In this model, the momentum conservation equation accounted for the effects of recoil pressure P_r (Eq. (7), [25]), surface tension F_γ (Eq. (8), [25]), Marangoni stress (Eq. (9), [25]) and buoyancy force F_b (Eq. (10), [20]) on the free surface, with the corresponding terms expressed as:

$$P_r = 0.54P_0 \exp\left(L_v \frac{T - T_b}{RTT_b}\right) \quad (7)$$

$$F_\gamma = \left(\gamma_0 + \frac{\partial \gamma}{\partial T}(T - T_m)\right) \cdot \frac{1}{R_c} \quad (8)$$

$$-\mu \frac{\partial v_t}{\partial n} = \frac{\partial \gamma}{\partial T} \frac{\partial T}{\partial \tau} \quad (9)$$

$$F_b = -\rho g \beta (T - T_m) \quad (10)$$

where P_0 is the standard atmospheric pressure, L_v denotes the latent heat of vaporization, T_b means the boiling point of the base metal, and R represents the universal gas constant. γ_0 indicates the surface tension of the base metal at the melting point, and T_m is the melting point, R_c denotes the radius of curvature. v_t represents the velocity vector along the tangential direction, n is the surface normal vector, and τ denotes the surface tangential vector. β means the thermal expansion coefficient.

To simulate the wire feeding process, the top surface was defined as a velocity inlet boundary. The movement of the base metal along the x direction was reproduced by setting the x boundaries as velocity inlet and outlet, relatively. A specific pressure boundary condition was applied to the bottom surface, while all other surfaces were assigned as wall boundary conditions.

3.4. Energy density distribution and laser absorption efficiency

In this study, the discretized energy was integrated over time to calculate the accumulated energy density distribution $U(x, y)$ in

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$$U(x, y) = \int_{z_{min}}^{z_{max}} \int_{T_1}^{T_2} \frac{E(x, y, z, t)}{d^3} dt dz \quad (11)$$

where z_{min} and z_{max} are the upper and lower bounds of the model in the z direction, t denotes the welding time, $T_1 - T_2$ represents the time range considered in the calculation, and d is the grid size.

The laser energy was classified into three pathways based on its destination in the computational domain: absorbed by the filler wire $\eta_{Filler\ wire}(t)$, the molten pool $\eta_{Molten\ pool}(t)$, and the solid base metal $\eta_{Solid\ base\ metal}(t)$. The sum of $\eta_{Filler\ wire}(t)$ and $\eta_{Molten\ pool}(t)$, which represents the total melting efficiency during the laser welding process, is defined as $\eta_{Total\ melting}(t)$, as shown in Eq. (12):

$$\left\{ \begin{array}{l} \eta_{Filler\ wire}(t) = \frac{\int \int \int_{\Omega_1} E(x, y, z, t) dV}{\int \int q(x, y) dx dy} \times 100\%, \quad \Omega_1 = \{V_S \geq 96\%\} \\ \eta_{Molten\ pool}(t) = \frac{\int \int \int_{\Omega_2} E(x, y, z, t) dV}{\int \int q(x, y) dx dy} \times 100\%, \quad \Omega_2 = \{V_S < 96\% \wedge T(x, y, z, t) \geq 1673\text{ K}\} \\ \eta_{Total\ melting}(t) = \eta_{Filler\ wire}(t) + \eta_{Molten\ pool}(t) \\ \eta_{Solid\ base\ metal}(t) = \frac{\int \int \int_{\Omega_3} E(x, y, z, t) dV}{\int \int q(x, y) dx dy} \times 100\%, \quad \Omega_3 = \{V_S < 96\% \wedge T(x, y, z, t) < 1673\text{ K}\} \end{array} \right. \quad (12)$$

where V_S is the fraction of filler wire, $T(x, y, z, t)$ denotes the temperature at (x, y, z, t) . Ω_1 represents the filler wire region, Ω_2 is the molten pool region, and Ω_3 corresponds to the solid base metal region, which is outside the molten pool. The filler wire region Ω_1 and molten pool region Ω_2 are delineated using a volume fraction threshold ($V_S \geq 96\%$). This threshold was determined through analysis of the filler wire region delineation and grid independence. As $V_S \geq 96\%$ provided a physically reasonable boundary consistent with high-speed images and minimized the grid dependence in laser absorption calculations, it was adopted as a robust threshold for laser absorption efficiency evaluation.

4. Results and discussion

4.1. Weld morphology under different wire offsets

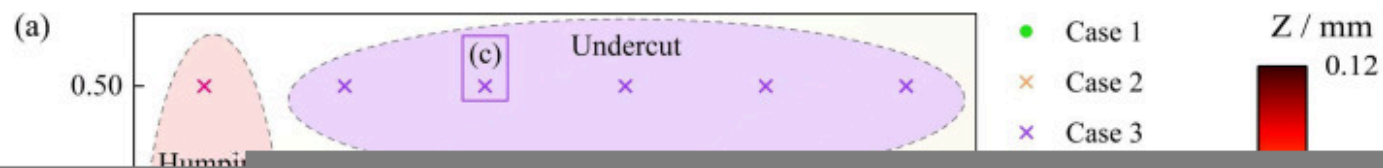
Fig. 3 shows the process window for wire offsets and the typical weld morphology corresponding to different wire offsets.

According to the GB/T 22085.1 and ISO 13919-1 standards, the weld morphology was categorized into four types: well-formed

Loading [MathJax]/jax/output/SVG/jax.js 5)), undercut (Case 3, [27]), and humping (Case 4, [28]), and each type was represented by a distinct symbol in Fig. 3(a). Additionally, the typical weld morphology is displayed by 3D point cloud data, and its height measurement

results are projected onto the XOY plane, as shown in Figs. 3(b)-(e). When $\Delta x = 0$ mm and $\Delta y = 0$ mm (Case 1), no apparent welding defects were found, and the weld seam was uniform on both the top and bottom surfaces. In the case of $\Delta x = 3$ mm and $\Delta y = 0$ mm (Case 2), significant underfill was observed on the top surface, particularly in the initial part of the weld seam (Fig. 3(c)). The underfill depth exceeded 0.07 mm, which corresponded to 5 % of the sheet thickness at a stringent qualitative level. When the filler wire was fed at $\Delta x = 0$ mm and $\Delta y = 0.5$ mm (Case 3), the transverse wire offset induced an asymmetric weld morphology and formation of undercut defects on the wire offset side, with a depth exceeding 0.07 mm (5 % of the sheet thickness), as shown in Fig. 3(d). A narrow ridge was observed between the undercut groove and the base metal. For $\Delta x = -2$ mm and $\Delta y = 0$ mm (Case 4), the humping defects occurred on both the top and bottom surfaces. The humping parts were higher than 0.14 mm (10 % of the sheet thickness), while the valley parts were deeper than 0.07 mm (5 % of the sheet thickness), as presented in Fig. 3(e). In general, the process window for a well-formed weld seam in Fig. 3(a) suggests that the weld morphology is more sensitive to the transverse wire offset than to the longitudinal wire offset. Furthermore, to improve the statistical robustness of this process window, we performed three independent replicates for each wire offset. Afterwards, the 3D point cloud data on the upper surface of the weld seam from the three replicates were averaged along the x direction to clearly reveal the defect characteristics. Although slight variations occurred due to minor fluctuations during the laser welding process, all replicates under the same wire offset exhibited identical defect types, which supported the reliability of this process window.

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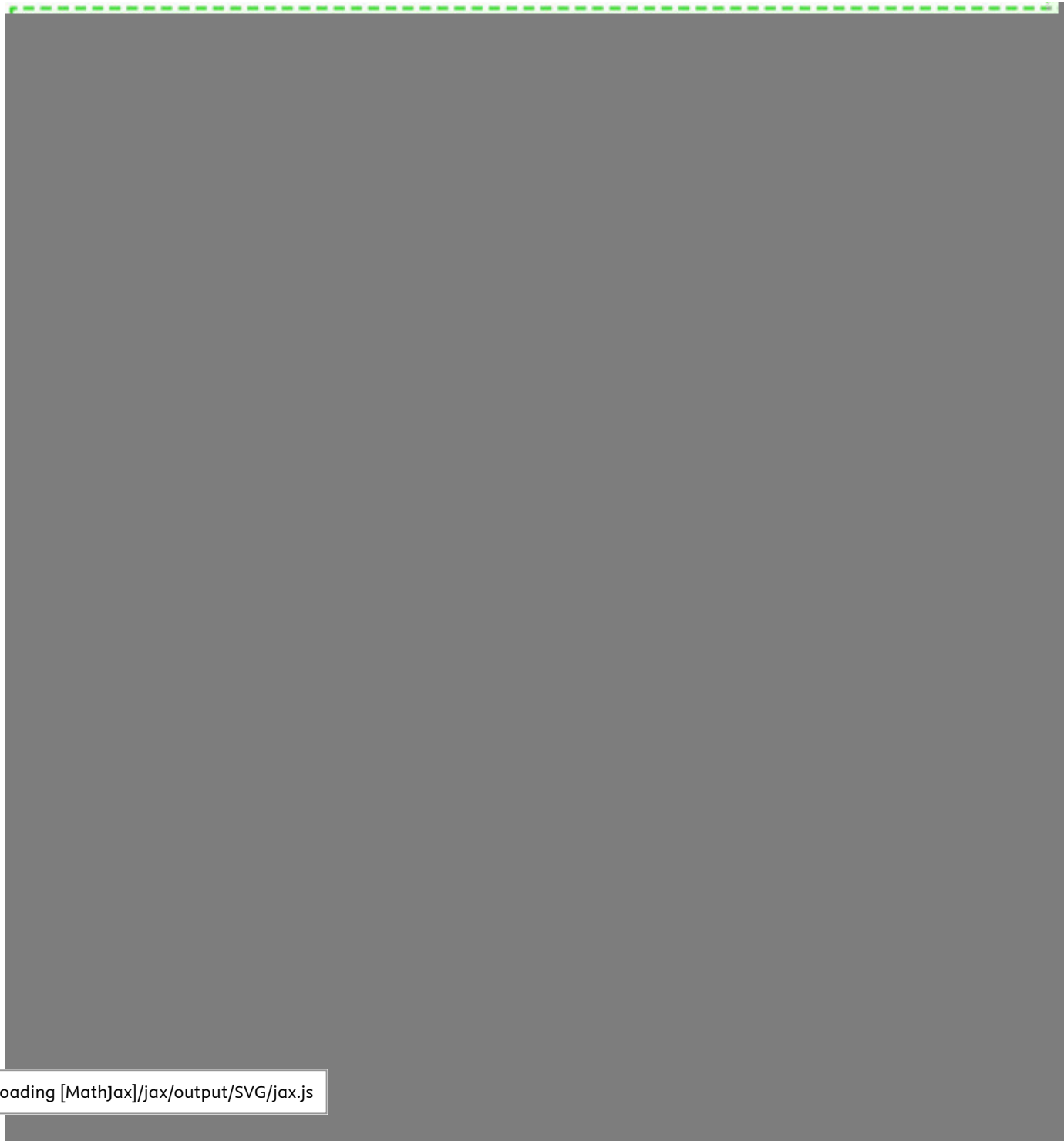
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Fig. 3. (a) Process window for the wire offsets. Typical weld morphology: (b) Well-formed (Case 1, $\Delta x = 0$ mm and $\Delta y = 0$ mm). (c) Underfill (Case 2, $\Delta x = 3$ mm and $\Delta y = 0$ mm, depth > 0.07 mm). (d) Undercut (Case 3, $\Delta x = 0$ mm and $\Delta y = 0.5$ mm, depth > 0.07 mm). (e) Humping (Case 4, $\Delta x = -2$ mm and $\Delta y = 0$ mm, depth > 0.07 mm and height > 0.14 mm).

4.2. Mass transfer behaviors and model validation

The variation in weld morphology under different wire offsets suggested that the mass transfer behaviors were not identical [29]. To elucidate the formation mechanisms of the different weld morphologies, the mass transfer behaviors and molten pool characteristics were analyzed using high-speed imaging and numerical simulation. Fig. 4 presents the mass transfer behaviors captured by the high-speed camera system and the numerical simulation results under the corresponding welding conditions. When $\Delta x = 0$ mm and $\Delta y = 0$ mm, the laser beam and the filler wire intersected at the leading edge of the molten pool, as shown in Fig. 4(a). The filler wire melted completely and flowed into the molten pool smoothly, thereby establishing a stable and continuous liquid bridge. A clockwise vortex was formed in the front region of the molten pool, and the molten metal in the lower region exhibited a backward flow. In this case, the molten pool was stable, and no significant defects were observed.

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Fig. 4. Mass transfer behaviors (a1-d1), and temperature distribution and flow patterns (a2-d2) under different wire offset positions. (a1, a2) $\Delta x = 0$ mm and $\Delta y = 0$ mm. (b1, b2) $\Delta x = 3$ mm and $\Delta y = 0$ mm. (c1, c2) $\Delta x = 0$ mm and $\Delta y = 0.5$ mm. (d1, d2) $\Delta x = -2$ mm and $\Delta y = 0$ mm.

For $\Delta x = 3$ mm and $\Delta y = 0$ mm, the filler wire was not fed into the molten pool in the initial stage of the laser welding process, leading to the formation of underfill defect, as shown in Fig. 4(b₁). In this stage, the laser welding process was equivalent to an
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 cess. With the filler wire feeding, it bent while gradually approaching the molten pool. Once the filler wire entered the molten pool, the direct irradiation of the laser beam melted the filler wire, establishing a stable liquid

bridge and eliminating the underfill defect. This process was similar to that with $\Delta x = 0$ mm and $\Delta y = 0$ mm. To reproduce the delay of wire feeding in the numerical model when $\Delta x = 3$ mm and $\Delta y = 0$ mm, the wire feeding speed was set to 0 before 0.27 s. The delay of 0.27 s was quantitatively determined from high-speed images in the case of $\Delta x = 3$ mm and $\Delta y = 0$ mm, which clearly showed that the filler wire required approximately 0.27 s to be transferred into the molten pool. Since the filler wire was not fed, the temperature around the keyhole was higher than that when $\Delta x = 0$ mm and $\Delta y = 0$ mm. Meanwhile, a strong downward flow was generated in the front region of the molten pool. However, the upward flow in the rear region of the molten pool was relatively weak, leading to an insufficient clockwise vortex. After the wire feeding speed increased to 1.5 m/min, the temperature around the keyhole decreased, thereby leading to a longer and more stable molten pool, as presented in Fig. 4(b₂). Meanwhile, the downward flow around the bottom of the keyhole was suppressed, while the upward flow in the rear region of the molten pool was promoted significantly, generating a stable and pronounced clockwise vortex.

In the case of $\Delta x = 0$ mm and $\Delta y = 0.5$ mm, although the filler wire was transferred in a stable liquid bridge mode, the molten pool was disturbed. With asymmetric wire feeding, a fraction of the molten filler wire flowed toward the wire offset side. Some of this molten metal overflowed the boundary of the molten pool, and a narrow ridge was formed adjacent to the undercut defect. The remaining molten filler wire was transferred to the leading edge of the molten pool. In the middle region of the molten pool, a relatively stronger counterclockwise vortex compared to the case when $\Delta x = 0$ mm, $\Delta y = 0$ mm was developed.

When the filler wire was fed at $\Delta x = -2$ mm and $\Delta y = 0$ mm, the droplet transfer mode dominated, which involved several stages, as shown in Fig. 4(d₁). Firstly, a droplet was formed at the wire tip. As the molten filler wire could not merge into the molten pool, the weld seam generated a valley part on the top surface. Once the droplet grew to a certain degree and contacted the molten pool, it spread on the top surface of the molten pool, forming the humping part. Subsequently, the droplet detached from the filler wire, and a new droplet began to develop. The temperature distribution and flow patterns are presented in Fig. 4(d₂). When the droplet was small, gravity was insufficient to detach it, and the molten metal flowed upward along the wire tip [30].

Fig. 5 compares the velocity distribution and flow patterns on the top surface of the molten pool under different wire offsets. For the case of $\Delta x = 0$ mm and $\Delta y = 0$ mm (Fig. 5(a)), the velocity distribution on the top surface was relatively symmetrical and uniform. With the filler wire feeding, the molten metal flowed to the low-temperature regions at the edge of the molten pool. Once the molten metal impinged on the boundary of the molten pool, it flowed backward and downward. Meanwhile, the molten metal in the rear region exhibited a sustained forward flow. When $\Delta x = 3$ mm and $\Delta y = 0$ mm, the velocity of the molten metal was significantly higher before wire feeding, as shown in Fig. 5(b). However, the length of the molten pool

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was only approximately 50 % of that when $\Delta x = 0$ mm and $\Delta y = 0$ mm. After wire feeding, the velocity in the front region of the molten pool decreased and became stable. In the upper region of the molten pool, a fraction of molten filler wire flowed backward along the boundary of the molten pool, which widened the molten pool. In the case of $\Delta x = 0$ mm and $\Delta y = 0.5$ mm, due to the asymmetric filler wire feeding, the velocity distribution appeared asymmetric, as shown in Fig. 5(c). As a fraction of molten filler wire was transferred to the rear and the center of the molten pool, the velocity of the molten metal in the front region was higher, particularly on the wire offset side. For $\Delta x = -2$ mm and $\Delta y = 0$ mm, as the energy absorbed by the droplet had not yet been transferred to the molten pool, the velocity was relatively lower, as shown in Fig. 5(d). After the droplet was integrated into the molten pool, the molten pool was drastically disrupted.

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Fig. 5. Velocity distribution and flow patterns on the top surface of the molten pool under different wire offsets. (a) $\Delta x = 0$ mm and $\Delta y = 0$ mm. (b) $\Delta x = 3$ mm and $\Delta y = 0$ mm. (c) $\Delta x = 0$ mm and $\Delta y = 0.5$ mm. (d) $\Delta x = -2$ mm and $\Delta y = 0$ mm.

Fig. 6 compares the weld profiles derived from experiments and simulations. The experimental weld profiles and weld widths matched well with those of the simulation. For the case of $\Delta x = 3$ mm and $\Delta y = 0$ mm, the cross-sections were taken from the region before the filler wire was transferred into the molten pool to ensure the reliability of the comparison. Additionally, the high-speed images matched well with the simulated results, as shown in Fig. 4, demonstrating the fidelity of the numerical models.

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Fig. 6. Comparisons of experimental and simulated weld profiles.

4.3. Effect of wire feeding on weld morphology

In the above experimental and simulated results, it was observed that when $\Delta x = 3$ mm and $\Delta y = 0$ mm, the stability of the

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improved due to the filler wire feeding. The stabilizing effect of the filler wire feeding was

consistent with experimental results in the previous work [12]. During the laser welding process, the stability of the molten pool strongly depended on its dynamic behaviors and keyhole evolutions, both of which were governed by the laser absorption efficiency and energy density [31]. Therefore, this section elucidated the interdependence among laser absorption efficiency, energy density distribution, and molten pool dynamics.

The laser absorption efficiency over time is presented in Fig. 7(a). Although the laser absorption efficiency of the molten pool decreased from 48.41 % to 45.73 % after wire feeding, the laser absorption efficiency of the filler wire (7.11 %) slightly elevated the total melting efficiency from 48.93 % to 52.38 %. This indicated that the filler wire acted not only as an additional mass source, but also as a regulator for energy density. Additionally, although the laser absorption efficiency of the solid base metal increased from 1.29 % before wire feeding to 1.73 % thereafter, it remained negligible in laser welding. Fig. 7(b) shows the accumulated energy density distribution. Most of the energy was concentrated within the weld seam. Owing to the different total melting efficiency before and after wire feeding, an increase in the accumulated energy density distribution occurred. The keyhole characteristics including keyhole X coordinate, keyhole depth, and keyhole opening area at a plane ($z = 1.25$ mm) are presented in Fig. 7(c). Before wire feeding, the keyhole underwent transient switching between the penetrated and blind states. After wire feeding, the filler wire feeding to the leading edge of the molten pool resulted in keyhole refilling. Therefore, the keyhole moved backward by approximately 0.2 mm, and its depth and opening area decreased.

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Fig. 7. Accumulated energy density distribution and keyhole characteristics when $\Delta x = 3$ mm and $\Delta y = 0$ mm. (a) Laser absorption efficiency over time. (b) Accumulated energy density distribution. (c) Keyhole characteristics over time. (d) Averaged total melting efficiency within different keyhole depth ranges. (e) Averaged total melting efficiency within different keyhole opening area ranges.

To quantify the relationship between the instantaneous keyhole characteristics and total melting efficiency, we calculated the time-averaged efficiency corresponding to various ranges of keyhole depth and opening area, as shown in Figs. 7(d)-(e). Before wire feeding (0.15–0.25 s), the total melting efficiency was highly sensitive to keyhole characteristics. The total melting efficiency decreased with increasing keyhole depth, as a deeper keyhole promoted multiple reflections of the laser beam [32]. However, this trend reversed after a fully penetrated keyhole was formed (decrease from 51.22 % to 46.24 %) due to the laser leakage through

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the bottom opening of the keyhole. For the keyhole opening area, a larger keyhole opening area facilitated the escape of the reflected laser beam from the keyhole, thereby reducing the total melting efficiency [33]. In contrast, after wire feeding (0.3–0.4 s), the total melting efficiency remained stable and largely insensitive to the keyhole depth and opening area. On one hand, the energy absorbed by the filler wire was relatively stable, which mitigated the effect of fluctuations in the energy absorbed by the molten pool. On the other hand, the keyhole maintained a relatively shallow and small state, keeping the total melting efficiency at a relatively comparable level.

Compared to the limited increase (3.45 %) in total energy density, the filler wire feeding had a greater effect on the dynamics and stability of the molten pool by significantly redistributing the spatial energy density. After integrating the energy density along different directions, the energy density along the x and z directions was averaged over time, as shown in Fig. 8. Before wire feeding, a characteristic peak in energy density ($4.80 \text{ J}\cdot\text{mm}^{-2}$) was observed at the laser incidence point. Since the laser beam reflected from the bottom and front wall of the keyhole to the rear wall, the average energy density on the rear wall was elevated ($0.77 \text{ J}\cdot\text{mm}^{-2}$). Additionally, a significant fraction of energy was concentrated at the bottom of the keyhole [34], where the energy density reached $5.57\text{--}6.59 \text{ J}\cdot\text{mm}^{-2}$. After wire feeding, the spatial energy density was redistributed markedly. A characteristic peak appeared around the upper surface of the base metal, corresponding to the energy directly absorbed by the filler wire ($7.57 \text{ J}\cdot\text{mm}^{-2}$). As the laser beam was reflected by the filler wire, the energy density on the rear wall of the keyhole was more dispersed and lower ($0.65 \text{ J}\cdot\text{mm}^{-2}$). Meanwhile, the energy density at the bottom of the keyhole ($4.54\text{--}4.95 \text{ J}\cdot\text{mm}^{-2}$) decreased compared to that before wire feeding. The variation in spatial energy density distribution before and after wire feeding resulted in significantly different driving forces and molten pool dynamics.

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Fig. 8. Average energy density distribution along the x direction (a1, b1) and z direction (a2, b2). (a1, a2) Before wire feeding. (b1, b2) After wire feeding. The keyhole profile illustrated in this figure corresponds to the keyhole characteristic in Fig. 7, and the average energy density distribution along the x direction is presented as a cloud map within the area under the keyhole profile.

Based on the analysis of laser absorption efficiency, keyhole characteristics, and spatial energy density distribution before and after wire feeding, the formation mechanism when $\Delta x = 3 \text{ mm}$ and $\Delta y = 0 \text{ mm}$ is illustrated in Fig. 9. Before wire feeding, the laser beam irradiated on the front wall of the keyhole and converged at the bottom of the keyhole through multiple reflections,

as shown in Fig. 9(a). Therefore, the energy density and temperature at the bottom of the keyhole were relatively higher, leading to a stronger recoil pressure. The intense recoil pressure drove the molten metal to flow downward [35], generating a fully penetrated keyhole, as presented in Fig. 9(b). Afterwards, a fraction of energy dissipated from the bottom of the keyhole, resulting in decreased total melting efficiency and temperature in the keyhole. As a result, the recoil pressure weakened, and the surface tension caused the keyhole to contract and form a blind keyhole (Fig. 9(c)) [36]. During this highly dynamic and rapid switching welding process, due to the formation of a deep and intermittent fully penetrated keyhole, the backward flow at the bottom of the molten pool was suppressed, and the high-temperature molten metal could not flow backward sufficiently. Therefore, the temperatures at the upper and lower surfaces of the molten pool were similar, failing to develop a pronounced upward Marangoni flow [37]. Consequently, the molten metal in the upper region of the molten pool could not be adequately supplied, ultimately leading to the formation of underfill defects.

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Fig. 9. Schematic diagram of formation mechanism of underfill defects when $\Delta x = 3$ mm and $\Delta y = 0$ mm. (a)-(c) Before wire feeding. (d) After wire feeding.

After wire feeding, the molten pool reached a relatively stable state, as shown in Fig. 9(d). On the leading edge of the molten pool, the molten filler wire was transferred into the molten pool, facilitating a pronounced downward flow along the front wall of the keyhole. The energy absorbed by the filler wire was transferred into the front wall of the keyhole, constituting 38.8 % of the total energy absorbed by the front wall. As this fraction of energy did not experience further reflection, the energy density on the rear wall of the keyhole was significantly lower than that without the filler wire feeding. Additionally, the filler wire feeding redirected the reflected laser beam. Although the laser beam reflected by the filler wire remained within the molten pool, it was dispersed on the rear wall of the keyhole, and the multiple reflected laser beams were no longer converged at the bottom of the keyhole. Therefore, in the keyhole, the recoil pressure was relatively weak, and the expansion tendency was suppressed. As a result, a relatively shallow keyhole was sustained, and the stability of the molten pool was maintained. Due to the filler wire feeding and the formation of a shallower keyhole, a liquid channel was formed under the bottom of the keyhole, which intensified the backward flow in the lower region of the molten pool significantly. This backward flow carried the high-

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the rear region of the molten pool. Consequently, the upward Marangoni flow was enhanced due to

the higher temperature difference between the upper and lower surfaces of the molten pool, generating a stable and pronounced clockwise vortex. Meanwhile, a fraction of molten filler wire flowed backward along the side wall of the keyhole. The integrated effect of these two mechanisms suppressed the formation of the underfill defects and led to a well-formed weld seam.

4.4. Formation mechanism of undercut defects with a transverse wire offset

Although the filler wire feeding contributes to the stability of the laser welding process, this stability requires precise alignment of the filler wire. In this study, it was found that the transverse wire offset had a more detrimental effect on weld morphology than the longitudinal wire offset. Counterintuitively, undercut defects tended to form on the wire offset side when a transverse wire offset was introduced. This finding suggested that the formation mechanism of the undercut defects was different from those in arc welding and hybrid laser-arc welding processes, where it was typically caused by insufficient energy and molten metal [38,39]. Therefore, we proposed a mechanistic model linking wire offset to the formation of the undercut defects in the laser welding with filler wire by analyzing the asymmetric energy density and fluid flow.

Quantitative analysis of the molten pool and keyhole dynamics reveals the basis for the asymmetry with a transverse wire offset, as shown in Fig. 10. The transverse wire offset resulted in significant asymmetry in both the molten pool and keyhole. As presented in Fig. 10(a), when $\Delta x = 0$ mm and $\Delta y = 0.5$ mm, the keyhole was predominantly asymmetric about the plane of incidence ($y = 4$ mm), with a displacement toward the opposite side of the wire. Therefore, the ratio of the keyhole opening areas on the upper and lower surfaces when $\Delta x = 0$ mm and $\Delta y = 0.5$ mm was significantly lower than that when $\Delta x = 0$ mm and $\Delta y = 0.5$ mm (Fig. 10(b)). Additionally, the velocity in the front and rear regions of the molten pool was also asymmetric, as shown in Figs. 10(c)-(d). A significant asymmetry in fluid flow occurred on the wire offset side, with the rear being lower and the front being higher than those on the opposite side, leading to asymmetric keyhole dynamics. Fig. 10(e) presents the keyhole dynamics in the YOZ plane over time. As the laser beam moved, the molten metal in the rear region of the molten pool flowed forward sustainably. On the wire offset side, the backward flow of molten metal from the front region was relatively vigorous, combined with the weak forward flow from the rear region, thereby preventing the filling of the keyhole and formation of undercut defects. Conversely, on the opposite side, a rapid and instantaneous forward flow stabilized and filled the keyhole sufficiently.

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Fig. 10. Molten pool and keyhole dynamics under different transverse wire offsets. (a) Keyhole profile at a plane ($z = 0.95$ mm). (b) Ratio of the keyhole opening area at $z = 0.95$ mm on the wire offset side ($y > 4$ mm) to that on the opposite side of the wire ($y < 4$ mm). (c) Velocity history in the front region and (d) in the rear region of the molten pool on two sides: the wire offset side ($9.8 \text{ mm} < x < 10.5 \text{ mm}$, $4.1 \text{ mm} < y < 4.3 \text{ mm}$, $1 \text{ mm} < z < 1.4 \text{ mm}$) and the opposite side ($9.8 \text{ mm} < x < 10.5 \text{ mm}$, $3.7 \text{ mm} < y < 3.9 \text{ mm}$, $1 \text{ mm} < z < 1.4 \text{ mm}$). (e) Keyhole dynamics in the YOZ plane over time.

To explain the asymmetry of the molten pool dynamics, the spatial energy density distribution in the molten pool was calculated. Figs. 11(a)–(b) compare the average energy density distributions along the y direction with ($\Delta x = 0$ mm and $\Delta y = 0.5$ mm) and without ($\Delta x = 0$ mm and $\Delta y = 0$ mm) a transverse wire offset. When the filler wire and the laser beam were aligned, the energy density was symmetric about the plane of incidence ($y = 4$ mm). However, when a transverse wire offset was introduced, a fraction of energy was transferred to the molten pool through the molten filler wire, causing a higher energy density on the wire offset side compared to its opposite side. Furthermore, the asymmetry in the average energy density distribution induced by the wire offset was also manifested in the energy density along the x direction, as shown in Figs. 11(c)–(d). On the opposite side of the wire (Fig. 11(c)), the energy density on the front wall of the keyhole was lower than that on the wire offset side (Fig. 11(d)), whereas the energy density on the rear wall of the keyhole was higher. As presented in Figs. 11(b) and 11(d), the energy absorbed by the filler wire was mainly concentrated on the wire offset side, leading to an increase in energy density on the front wall of the keyhole.

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Fig. 11. Average energy density along the y direction (Energy density: $0\text{--}5 \text{ J}\cdot\text{mm}^{-2}$). (a) $\Delta x = 0$ mm and $\Delta y = 0$ mm. (b) $\Delta x = 0$ mm and $\Delta y = 0.5$ mm. Average energy density along the x direction when $\Delta x = 0$ mm and $\Delta y = 0.5$ mm (Energy density: $0\text{--}2 \text{ J}\cdot\text{mm}^{-2}$). (c) $0 \text{ mm} < y < 4 \text{ mm}$. (d) $4 \text{ mm} < y < 8 \text{ mm}$.

To illustrate the cause of energy density asymmetry, the energy propagation and force analysis under different transverse wire offsets are depicted in Fig. 12. When $\Delta x = 0$ mm and $\Delta y = 0$ mm, the keyhole exhibited a symmetric opening. Additionally, the front wall of the keyhole remained nearly perpendicular to the welding direction. Therefore, the laser beam reflected onto the

relatively symmetric and concentrated. However, when a transverse wire offset occurred ($\Delta x = 0$ mm and $\Delta y = 0.5$ mm, Fig. 12(b)), the asymmetric filler wire feeding induced an acute angle between the front wall of the

keyhole and the welding direction on the wire offset side. After the laser beam underwent its first reflection, the laser beam tended to irradiate on the rear wall of the keyhole on the opposite side of the wire. This phenomenon was also indicated in the asymmetric laser absorption efficiency of the solid base metal. The average laser absorption efficiency of Sheet 1 was 1.42 %, while that of Sheet 2 was only 0.72 %. This significant difference suggested that the majority of the laser beam was reflected to the opposite side of the wire. Consequently, a higher energy density and elevated temperature occurred on the rear wall of the keyhole opposite the wire, which played a crucial role in determining the magnitude of the driving forces.

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Fig. 12. Schematic diagram of energy propagation and force analysis. (a) $\Delta x = 0$ mm and $\Delta y = 0$ mm in the XOY plane. (b) $\Delta x = 0$ mm and $\Delta y = 0.5$ mm in the XOY plane. (c) $x = 0$ mm and $\Delta y = 0.5$ mm in the XOZ plane ($y < 4$ mm) at different times (from t_1 to t_2). (d) $x = 0$ mm and $\Delta y = 0.5$ mm in the XOZ plane ($y > 4$ mm) at different times (from t_1 to t_2).

During the laser welding process, the expansion tendency of the keyhole was primarily driven by the recoil pressure P_r (Eq. (7)), while the contraction tendency of the keyhole was governed by the surface tension F_γ (Eq. (8)) [36]. As the Marangoni stress acted tangentially along the front and rear walls of the keyhole, its contribution to the keyhole evolution along the welding direction was relatively weak [40]. To elucidate the effect of the driving force in the molten pool, a semi-quantitative force analysis was performed on the rear wall of the keyhole on the wire offset side ($y > 4$ mm) and its opposite side ($y < 4$ mm), as shown in Figs. 12(c)-(d). Compared to the wire offset side (3000 K, 1.7×10^4 Pa), the temperature on the opposite side of the wire was higher, resulting in a higher recoil pressure (3100 K, 3.1×10^4 Pa). Unlike the exponential increase of the recoil pressure with the elevated temperature, surface tension increased linearly with the elevated temperature. Thus, a minor temperature variation (~ 100 K) above 3000 K did not significantly influence the surface tension. However, when the keyhole was under a quasi-steady state, the surface tension was approximately equal to the recoil pressure [41]. Therefore, in this case, the surface tension on the wire offset side was only half of that on its opposite side. Based on the keyhole profile in Fig. 10(a), it could be inferred that this difference was caused by the variations in the radius of curvature. As the laser beam moved forward, the recoil pressure on the rear wall of the keyhole decreased rapidly as the temperature dropped, while the surface tension remained relatively high.

Therefore, on the opposite side of the wire, the refilling effect of the keyhole was significantly stronger than that on the wire

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offset side. This imbalance in the refilling of the keyhole resulted in insufficient molten metal during solidification, thereby forming undercut defects on the wire offset side, as mentioned in Fig. 10(e).

To further support the proposed mechanism that the undercut formation was dominated by the asymmetric energy deposition between the front and rear walls, a predictive model was established a modulated Gaussian heat source, while keeping all other physical conditions (wire feeding, Marangoni coefficient, recoil pressure model, etc.) identical to the baseline case ($\Delta x = 0$ mm and $\Delta y = 0.5$ mm). During Stage I (0–0.3 s), the heat source remained a Gaussian distribution. During Stage II (0.3–0.6 s), the laser power density on the wire offset side ($y > 4$ mm) was increased by 10 %, and that on its opposite side ($y < 4$ mm) was decreased by 10 %. This modulation strategy guaranteed the conservation of the total laser power. During Stage III (0.6–0.9 s), the above modulation strategy was inverted, with a 10 % decrease on the wire offset side and a 10 % increase on its opposite side. The modulation of laser power density significantly influenced the formation of undercut defects, with the undercut defects aggravated in Stage II but suppressed in Stage III. To elucidate the mechanism, we calculated the ratio of keyhole opening area and velocity in the front region of the molten pool between the wire offset side and its opposite side. In Stage II, the increased laser power density on the wire offset side promoted the backward flow, expanding the keyhole on this side. However, the forward flow on the rear wall of the keyhole, which was important for the formation of undercut defects, remained largely unchanged. This imbalance in fluid flow hindered the refilling of the keyhole, thereby aggravating the undercut defects. Conversely, in Stage III, the lower energy density on the wire offset side weakened the backward flow, contracting the keyhole and promoting complete refilling of the keyhole, which effectively suppressed the formation of undercut defects Fig. 13.

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Fig. 13. Effect of laser power density distributions on the formation of undercut defects. The pale orange and pale green dashed lines represent the average ratio of keyhole opening area and velocity in the front for the baseline case.

5. Conclusions

Using an integrated experimental and numerical approach, this study establishes a stable process window for wire offsets in laser welding with filler wire and elucidates the formation mechanisms of welding defects outside this window. The link between energy density and molten pool dynamics is established, revealing the role of the filler wire as both a mass source and a regulator

for energy density. Additionally, a new mechanistic model is proposed for the formation of undercut defects. The main conclusions are as follows:

- (1) The longitudinal wire offset (Δx) governs process stability by determining the mass transfer mode. A stable liquid bridge transfer mode is established with precise alignment, ensuring well-formed weld morphology. A positive longitudinal offset ($\Delta x = 3 \text{ mm}$) initially induces underfill due to the delay of wire feeding. However, once a stable liquid bridge is established, the underfill defect is eliminated. Conversely, a negative offset ($\Delta x = -2 \text{ mm}$) dominates the unstable droplet transfer mode, leading to the formation of humping defects.
- (2) The filler wire feeding acts as a regulator, which redistributes energy density and improves process stability. It absorbs a fraction of the incident laser energy (7.11 %) and transfers it directly to the molten pool, thereby mitigating vigorous downward flow and suppressing the formation of a fully penetrating keyhole. A longer, more stable molten pool is generated after wire feeding, eliminating the formation of underfill defects.
- (3) The transverse wire offset (Δy) has a more detrimental effect on weld morphology than the longitudinal offset. A minor transverse wire offset ($\Delta y = 0.5 \text{ mm}$) disturbs the symmetry of energy density and molten pool dynamics. Although the front wall of the keyhole on the offset side absorbs additional energy from the molten filler wire, the rear wall on the offset side is irradiated with less reflected laser energy. This asymmetry results in insufficient refilling of the keyhole and causes the formation of undercut defects on the wire offset side.

CRedit authorship contribution statement

Zihan Li: Writing – original draft, Validation, Software, Methodology, Investigation, Conceptualization. **Fang Li:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis. **Zhengxian Li:** Supervision, Resources, Investigation. **Yilin Chen:** Supervision, Resources, Investigation. **Gang Ruan:** Visualization, Validation, Data curation. **Jiacheng Huang:** Visualization, Data curation. **Lin Wang:** Methodology, Investigation. **Chen Shen:** Supervision, Resources, Project administration. **Xueming Hua:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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